

CS 224R Proposal

Ethan Hersch Ryan D’Cunha Abhinav Chinta
Department of Computer Science
Stanford University
{ehersch, rdcunha, achinta}@stanford.edu

April 30, 2026

1 Objective

Deep reinforcement learning (RL) training pipelines are constrained by the trade-off between sample efficiency and optimization stability. Off-policy algorithms, such as Soft Actor-Critic (SAC), excel at early exploration and are highly sample efficient. However, they suffer from Q-value overestimation and training instability. Conversely, on-policy algorithms, such as Proximal Policy Optimization (PPO), provide stable convergence but are sample inefficient.

This project investigates the transitional dynamics of algorithm sequencing—transitioning an agent from off-policy exploration (SAC) to on-policy stabilization (PPO) mid-training. We plan to investigate a compute allocation strategy that accelerates convergence without sacrificing performance. Investigating these algorithmic handoffs is crucial for foundation model alignment as on-policy RLHF creates a compute bottleneck for frontier models. Demonstrating that sequenced objective functions yield superior optimization trajectories under strict compute constraints could inform next-generation alignment pipelines.

2 Related Work

Deep reinforcement learning algorithms often trade off sample efficiency against optimization stability. Trust Region Policy Optimization (TRPO) and Proximal Policy Optimization (PPO) constrain policy updates to improve training stability, making them standard on-policy policy-gradient baselines [16, 15]. In contrast, Soft Actor-Critic (SAC) is an off-policy maximum-entropy actor-critic method that reuses data through a replay buffer and typically improves sample efficiency [6]. However, off-policy actor-critic methods can suffer from value estimation error and overestimation, motivating methods such as TD3 that explicitly reduce function approximation error [3]. These differences motivate our study of whether SAC is better suited for early exploration while PPO is better suited for later-stage stabilization.

Several prior methods attempt to combine the benefits of on-policy and off-policy reinforcement learning. ACER improves sample efficiency by incorporating experience replay into actor-critic learning [18]. Q-Prop uses an off-policy critic as a control variate for policy-gradient estimation [5], while PGQL and Interpolated Policy Gradient combine policy-gradient updates with Q-learning or off-policy gradient estimates [12, 4]. More recent work such as P3O and GePPO studies how PPO-style policy optimization can reuse off-policy samples while retaining stability benefits [2, 13]. These approaches design hybrid objectives, whereas we study explicit algorithmic handoffs between distinct algorithms.

Staged training pipelines are also common when demonstrations, offline data, or teacher policies are available. Deep Q-learning from Demonstrations uses demonstration data to accelerate early learning [7]. AWAC and IQL show that offline datasets can initialize policies that are later improved with online reinforcement learning [11, 9]. Kickstarting Deep Reinforcement Learning uses pretrained teacher agents to guide student agents during training [14]. These works show that transitions between training regimes can improve learning, but they mainly study transitions from demonstrations, offline data, or teacher policies into RL rather than the ordering of two online RL algorithms.

Curriculum learning and adaptive training schedules provide another perspective on sequencing. Curriculum learning studies how ordering training experience can improve convergence and generalization [17], and teacher-student curriculum learning dynamically selects tasks based on learner progress [10]. Population Based Training discovers hyperparameter schedules under a fixed compute budget [8], while learning-to-optimize methods train neural networks to control optimization procedures [1]. These methods adapt tasks, hyperparameters, or optimizers during training; in contrast, our project studies whether switching the RL algorithm itself improves sample efficiency, stability, and final performance under controlled compute budgets.

3 Technical Approach

We propose a systematic study of algorithm ordering and switching strategies in RL. Our central hypothesis is that different algorithms specialize in different stages of learning (exploration, stabilization, or fine-tuning, and that sequencing them appropriately can improve training performance. The novel technical contribution lies in formalizing the algorithmic handoff and designing internal adaptive switching triggers. All experiments will be conducted using continuous-control benchmark environments from the MuJoCo suite (Hopper-v4, Walker2d-v4) and a sparse-reward gridworld environment.

We focus on three complementary experiments.

- **Experiment 1: Fixed Algorithm Ordering:** We will benchmark fixed-schedule pipelines (SAC $\rightarrow$ PPO) against single-algorithm baselines (pure SAC, PPO) to evaluate early sample efficiency and final asymptotic reward
- **Experiment 2: Adaptive Algorithm Switching:** Instead of relying reward

plateaus (that may lag), we will evaluate internal metrics for the algorithm switch. Specifically, SAC to PPO based on a threshold for the policy’s KL divergence, indicating off-policy saturation

- **Experiment 3:** Compute-Constrained Scheduling: To fairly evaluate the pipelines as compute-allocation strategies, training budgets will be constrained by a fixed number of total gradient updates/FLOPs, rather than wall-clock time or environment interactions. This isolates whether spending compute on SAC early in training yields a superior optimization trajectory compared to PPO.

4 Expected Contributions

Ryan will scope out more methods existing literature, implement baselines, and lead experiment 1. Ethan will work on adaptive algorithm switching in experiment 2 and explore potential benchmarking literature for other experimentation environments. Abhinav will work on compute constrained scheduling and experiment with the strategies, training budgets, and metrics collected.

References

- [1] Marcin Andrychowicz et al. *Learning to learn by gradient descent by gradient descent*. 2016. arXiv: 1606.04474 [cs.NE]. URL: <https://arxiv.org/abs/1606.04474>.
- [2] Rasool Fakoor, Pratik Chaudhari, and Alexander J. Smola. *P3O: Policy-on Policy-off Policy Optimization*. 2019. arXiv: 1905.01756 [cs.LG]. URL: <https://arxiv.org/abs/1905.01756>.
- [3] Scott Fujimoto, Herke van Hoof, and David Meger. *Addressing Function Approximation Error in Actor-Critic Methods*. 2018. arXiv: 1802.09477 [cs.AI]. URL: <https://arxiv.org/abs/1802.09477>.
- [4] Shixiang Gu et al. *Interpolated Policy Gradient: Merging On-Policy and Off-Policy Gradient Estimation for Deep Reinforcement Learning*. 2017. arXiv: 1706.00387 [cs.LG]. URL: <https://arxiv.org/abs/1706.00387>.
- [5] Shixiang Gu et al. *Q-Prop: Sample-Efficient Policy Gradient with An Off-Policy Critic*. 2017. arXiv: 1611.02247 [cs.LG]. URL: <https://arxiv.org/abs/1611.02247>.
- [6] Tuomas Haarnoja et al. *Soft Actor-Critic Algorithms and Applications*. 2019. arXiv: 1812.05905 [cs.LG]. URL: <https://arxiv.org/abs/1812.05905>.
- [7] Todd Hester et al. *Deep Q-learning from Demonstrations*. 2017. arXiv: 1704.03732 [cs.AI]. URL: <https://arxiv.org/abs/1704.03732>.
- [8] Max Jaderberg et al. *Population Based Training of Neural Networks*. 2017. arXiv: 1711.09846 [cs.LG]. URL: <https://arxiv.org/abs/1711.09846>.

- [9] Ilya Kostrikov, Ashvin Nair, and Sergey Levine. *Offline Reinforcement Learning with Implicit Q-Learning*. 2021. arXiv: 2110.06169 [cs.LG]. URL: <https://arxiv.org/abs/2110.06169>.
- [10] Tambet Matiisen et al. *Teacher-Student Curriculum Learning*. 2017. arXiv: 1707.00183 [cs.LG]. URL: <https://arxiv.org/abs/1707.00183>.
- [11] Ashvin Nair et al. *AWAC: Accelerating Online Reinforcement Learning with Offline Datasets*. 2021. arXiv: 2006.09359 [cs.LG]. URL: <https://arxiv.org/abs/2006.09359>.
- [12] Brendan O’Donoghue et al. *Combining policy gradient and Q-learning*. 2017. arXiv: 1611.01626 [cs.LG]. URL: <https://arxiv.org/abs/1611.01626>.
- [13] James Queeney, Ioannis Ch. Paschalidis, and Christos G. Cassandras. *Generalized Proximal Policy Optimization with Sample Reuse*. 2021. arXiv: 2111.00072 [cs.LG]. URL: <https://arxiv.org/abs/2111.00072>.
- [14] Simon Schmitt et al. *Kickstarting Deep Reinforcement Learning*. 2018. arXiv: 1803.03835 [cs.LG]. URL: <https://arxiv.org/abs/1803.03835>.
- [15] John Schulman et al. *Proximal Policy Optimization Algorithms*. 2017. arXiv: 1707.06347 [cs.LG]. URL: <https://arxiv.org/abs/1707.06347>.
- [16] John Schulman et al. *Trust Region Policy Optimization*. 2017. arXiv: 1502.05477 [cs.LG]. URL: <https://arxiv.org/abs/1502.05477>.
- [17] Xin Wang, Yudong Chen, and Wenwu Zhu. *A Survey on Curriculum Learning*. 2021. arXiv: 2010.13166 [cs.LG]. URL: <https://arxiv.org/abs/2010.13166>.
- [18] Ziyu Wang et al. *Sample Efficient Actor-Critic with Experience Replay*. 2017. arXiv: 1611.01224 [cs.LG]. URL: <https://arxiv.org/abs/1611.01224>.